Earth as a System: Energy, Matter, and Life

Class 9 Science (NCERT Exploration) | Chapter 13 | Quick Revision Notes

1. The Earth System and its Five Spheres

Life on Earth runs on a constant flow of **energy** (mainly from the Sun, plus Earth's hot interior and chemical reactions) and **matter**. Earth works as **one system** of interacting 'spheres':

Sphere	What it includes	Examples
Geosphere	Solid rocks, soil, landforms, Earth's interior	Deccan plateau, Thar desert
Hydrosphere	Liquid water	Oceans, rivers (Ganga-Brahmaputra), lakes, groundwater
Cryosphere	Solid water (ice and snow)	Himalayan glaciers, Ladakh snow, polar ice caps
Atmosphere	Air around the Earth	Air in mountains and forests
Biosphere	All living organisms and habitats	Mangroves, forests, farms, plankton, coral reefs

Key idea: a disturbance in one sphere changes the others
Less snowfall (cryosphere) -> less lake water (hydrosphere) -> less grass for sheep (biosphere).
Warmer Arabian Sea -> more evaporation -> fluctuating southwest monsoon -> floods in some regions, drought in others.
Warmer atmosphere -> melting glaciers and polar ice -> sea-level rise -> threat to coastal cities and habitat loss.

2. Uneven Heating of the Earth (13.1)

Solar radiation and the electromagnetic spectrum

- Sun's energy reaches Earth as **electromagnetic (EM) waves**; they travel through vacuum at the speed of light, 3 x 10^8 m/s. (Sound is a *mechanical* wave and needs a medium.)
- **Electromagnetic spectrum** (high to low frequency): gamma rays, X-rays, UV, visible, IR, microwaves, radio waves.
- About **99%** of solar energy lies in the **UV, visible and IR** range. Gamma and X-rays are mostly filtered by the upper atmosphere; microwaves and radio waves carry too little energy to warm Earth.

Radiation	Role
UV (100-400 nm)	Mostly absorbed by the ozone layer; protects life. Excess harms eyes and skin and raises cancer risk.
Visible light	Drives photosynthesis (primary source of food); partly warms land and water.
Infrared (IR)	Warms Earth's surface, which re-radiates heat; part of it is trapped by greenhouse gases.

Important terms

- **Insolation**: the Sun's radiation that reaches Earth's surface; it warms the surface and atmosphere.
- **Solar constant**: average solar energy per unit time per unit area, perpendicular to the Sun's rays, at the top of the atmosphere = about **1.4 kW/m^2** (1400 J s^{-1} m^{-2}).
- Maximum insolation at the surface (clear sky) is about **1 kW/m^2**, lower than the solar constant because gases, clouds and dust absorb or scatter some radiation.
- India lies in the tropical/sub-tropical zone and gets abundant sunlight; insolation drives the **southwest monsoon** and gives great solar-power potential.
- **Anna Mani** mapped insolation over India; with S. Rangarajan she published *Solar Radiation Over India* (1982), the first insolation atlas.

Example 13.1
Energy in 1 hour on 1 m^2 at 1 kW/m^2: E = intensity x area x time = 1000 x 1 x 3600 = **3.6 x 10^6 J** (about 1 unit of household electricity; enough to melt 5 kg of ice and heat the water to 100 °C).

Albedo (13.1.1)

- **Albedo** = fraction of solar radiation reflected by a surface (Latin: 'whiteness').
- **High albedo** = reflects more, stays cool. **Low albedo** = absorbs more, heats faster.

Material	Albedo	Effect
Snow	0.80 - 0.90	Very high; keeps polar regions cold
Ice	0.50 - 0.70	High
Crushed rock	0.25 - 0.30	Moderate
Black soil, ocean water	Low	Absorb more radiation, relatively warmer

- Land heats faster than water; dark surfaces (dark roads, dark clothes) heat more than light ones.
- **Urban heat island effect**: cities are warmer than rural areas (especially at night) because steel, concrete and asphalt absorb and re-radiate heat, while rural areas have vegetation that cools through shade and transpiration. It raises air-conditioning demand. Mud-and-wood houses stay cooler as they re-radiate less.

Latitude and Earth's shape (13.1.2)

- Earth is spherical, so the Sun's rays strike different latitudes at different angles.
- At the **equator** radiation is concentrated over a **small area** (warm all year); at the **poles** the same radiation is spread over a **larger area** (much colder). This creates the equator-pole temperature difference.
- Earth's tilt and revolution cause seasons and changing day length. Uneven heating drives **global winds and ocean currents**.

Role of the atmosphere (13.1.3)

- Held by gravity; about **78% nitrogen, 21% oxygen**, plus argon, CO_2, water vapour, etc.
- **Two protective roles:** (1) partly absorbs incoming radiation (ozone blocks UV; clouds and gases absorb some sunlight); (2) traps outgoing heat. Earth's surface re-radiates IR, and greenhouse gases (CO_2, CH_4, water vapour) absorb it. Without this, Earth would be too cold for life.
- Excess CO_2 from human activity strengthens the greenhouse effect and causes **global warming**.
- **Venus** is hotter than Mercury (though farther from the Sun) because of its atmosphere and an uncontrolled greenhouse effect.

Layer	Height	Features
Troposphere	0 - 12 km (avg.)	Nearly all weather (winds, storms, rain). Heated from the surface; temperature **falls** with height (~6.5 °C per km). Tallest over the equator, lowest over the poles.
Stratosphere	12 - 50 km	Contains the **ozone layer**, which absorbs UV; temperature **rises** with height. Little vertical mixing, so weather stays in the troposphere.
Mesosphere, thermosphere, exosphere	Above 50 km	Minor role in surface climate. Outer space begins at about 100 km.

Ozone layer
Acts as a UV shield. CFCs (refrigerators, aerosols) caused the Antarctic **ozone hole**. The **Montreal Protocol** cut CFC use and the layer is slowly recovering, which shows the power of international cooperation. (Ground-level ozone, formed from vehicle emissions, is harmful smog.)

3. Uneven Heating Causes Wind and Ocean Currents (13.2)

Wind = movement of air from **high pressure to low pressure**. Pressure differences arise mainly from uneven heating of Earth's surface.

Local winds

	Valley breeze (day)	Mountain breeze (night)
Cause	Slopes facing the Sun heat faster than the valley floor	Slopes lose heat faster and cool; valley floor stays warmer

	Valley breeze (day)	Mountain breeze (night)
Air movement	Warm air over slopes rises (low pressure); cooler valley air flows **up** the slope	Cool, dense air over slopes flows **down** into the valley

Seen in Shimla, Dehradun and Himalayan valleys. They influence weather and agriculture, and help regulate temperature and moisture, supporting soil and crop health.

Planetary winds

- **Equator:** intense heating, warm air rises, so an **equatorial low-pressure belt** forms. Air moves poleward at height, cools, and sinks near **30° N/S**, forming **sub-tropical high-pressure belts**. Surface air flows back to the equator.
- Part of the sinking air moves poleward and rises near **60° N/S**, meeting cold polar air, forming **sub-polar low-pressure belts**.
- At the **poles (90°)** cold, dense air sinks, forming **polar high-pressure belts**; air flows to the sub-polar belts.
- **Earth's rotation deflects winds**: to the **right** in the Northern Hemisphere and to the **left** in the Southern Hemisphere, so winds follow curved paths.

Ocean currents

- **Ocean currents** = continuous movement of large masses of ocean water.
- **Causes:** planetary winds dragging surface water (friction); differences in temperature and salinity; Earth's rotation; distribution of landmasses.
- Warm equatorial water flows along the surface towards the poles; colder, denser water returns to the equator at depth. **Low salinity** water is less dense and stays near the surface; **high salinity** water is denser and sinks.
- **Gyres**: large circular patterns formed by rotation-deflected currents; **clockwise** in the Northern Hemisphere, **counter-clockwise** in the Southern Hemisphere. Continents block and redirect currents.
- **Importance:** carry heat from equator to poles (reducing temperature differences), regulate climate, transport nutrients that support marine ecosystems, and support trade. Example: the warm **North Atlantic Drift** (extension of the **Gulf Stream**) keeps many NW European ports ice-free in winter.
- **IITM Pune** uses coupled atmosphere-ocean-land-ice computer models to simulate the Indian monsoon and study how global warming may change it.

4. Biogeochemical Cycles (13.3)

The **biogeochemical cycle** is the cyclic movement of matter and energy between **abiotic** (non-living) and **biotic** (living) components. It keeps nutrients such as carbon, nitrogen and oxygen available to life and helps ecosystems recover from disturbance and stay balanced.

Water cycle

- Water evaporates from oceans, rivers and lakes, **condenses** into clouds, returns as **precipitation** (rain, hail, snow), runs off to the ocean, or **infiltrates** the soil to become groundwater. Plants add water by **transpiration**. Water also dissolves minerals and carries nutrients to the oceans.
- **Climate change effects:** a warmer atmosphere holds more moisture, giving heavier rain in some areas (intensified monsoons) and droughts elsewhere. Melting glaciers raise sea levels (threat to Mumbai, Chennai). Intense bursts of rain cause more run-off and soil erosion and less infiltration, so groundwater recharge falls and farming becomes harder in dry months.
- The water cycle links the cryosphere, hydrosphere, atmosphere, geosphere and biosphere.

Carbon cycle

- Carbon is the backbone of life (proteins, carbohydrates, fats, DNA). It moves between the atmosphere (CO_2), biosphere, geosphere (carbonate rocks, fossil fuels) and hydrosphere (dissolved CO_2, marine shells).
- **Fast cycle (days to years):** photosynthesis removes CO_2; respiration and decomposition return it.
- **Slow cycle (millions of years):** buried dead organisms become coal, oil and gas; burning them releases CO_2 very quickly.
- **Ocean:** absorbs atmospheric CO_2 to form carbonate and bicarbonate ions, used by phytoplankton for photosynthesis and by some organisms for shells. Dead organisms sink and store carbon for a long time.
- Carbon is ~49% of the dry weight of living organisms; **71%** of global carbon is in the oceans; the atmosphere holds only about **1%**.

- **Human impact:** burning fossil fuels and deforestation raised atmospheric CO_2 by about 35% since 1960 (about 315 to 420 ppm; the Keeling curve). Result: stronger greenhouse effect, global warming, melting ice, sea-level rise and extreme weather.

Nitrogen cycle

- Nitrogen is needed for proteins and nucleic acids. The atmosphere is the largest reservoir, but N_2 is non-reactive and plants cannot use it directly.

Step	What happens	Organisms
Nitrogen fixation	Atmospheric N_2 to ammonia (NH_3); lightning also fixes nitrogen oxides	*Rhizobium* (legume root nodules), *Azotobacter* (soil)
Nitrification	Ammonia to nitrite (NO_2^-) to nitrate (NO_3^-)	*Nitrosomonas* (to nitrite), *Nitrobacter* (to nitrate)
Assimilation	Plants absorb nitrogen compounds; animals get nitrogen by eating plants or animals	Plants, animals
Ammonification	Organic matter from dead organisms and waste is broken down, returning ammonia to soil	Decomposers (bacteria, fungi)
Denitrification	Nitrates converted back to N_2 gas, completing the cycle	*Pseudomonas*

Ready to Go Beyond: Haber-Bosch process
Early 1900s method of making ammonia from atmospheric nitrogen ('Bread from Air'); source of most fertilisers and the Green Revolution. More than half the nitrogen atoms in the human body come from it. But it is energy-intensive (~1-2% of global energy) and fertiliser overuse degrades soil and water.

Oxygen cycle

- About **21%** of air is O_2. It is also found in the Earth's crust (metal oxides, minerals) and in air as CO_2.
- **Used up by:** respiration (plants and animals), combustion of fuels, oxide formation. **Restored by:** mainly **photosynthesis**.

5. Human Impact on Earth's Processes (13.4)

Human activity	Consequence
Burning fossil fuels, deforestation	Excess CO_2, greenhouse warming; forests and oceans become saturated carbon sinks. Warmer oceans absorb less CO_2.
Excess CO_2 absorbed by oceans	Sea water becomes more **acidic**, threatening plankton and coral reefs.
Overuse of fertilisers	Nitrates enter rivers and lakes, causing algal blooms that deplete oxygen and kill fish: **eutrophication**.
Deforestation	Less photosynthesis and transpiration (lower local rainfall), changed albedo, more soil erosion, habitat and biodiversity loss.
Vehicle emissions	Ground-level smog and ozone, harmful to health.

Solutions

- Local action plus global cooperation. The **Montreal Protocol** has been successful (ozone recovery); the **Kyoto Protocol** and **Paris Agreement** (CO_2 reduction) have been less successful.
- Conserve energy and resources; switch to **renewables** (solar, wind); plant trees; save water; practise sustainable farming; reduce, reuse, recycle.
- **Mission LiFE** (Lifestyle for Environment): India-led initiative introduced at the UN Climate Change Conference, 2021, promoting mindful, eco-friendly lifestyles.

6. Scientists Mentioned

Scientist / Institution	Contribution
Anna Mani	Mapped solar insolation across India; *Solar Radiation Over India* (1982) with S. Rangarajan.

Scientist / Institution	Contribution
K. R. Ramanathan	Measured ozone at 18,000 ft in the Himalayas (1934); found levels lower than expected; early monsoon forecasting.
IITM, Pune	Coupled atmosphere-ocean-land-ice models to simulate the Indian monsoon.

7. Quick Revision Points

- Sun's EM radiation is the primary energy source; ~99% is UV + visible + IR.
- Solar constant ~1.4 kW/m^2; surface insolation ~1 kW/m^2 (clear sky).
- High albedo = cool (snow, ice); low albedo = warm (black soil, ocean).
- Equator gets concentrated radiation; poles get spread-out radiation; hence the temperature difference.
- Weather happens in the troposphere; the stratosphere holds the ozone layer.
- Atmosphere (a) absorbs part of incoming radiation and (b) traps outgoing heat (greenhouse effect).
- Uneven heating gives local winds, planetary winds (deflected right in N. Hemisphere, left in S. Hemisphere) and ocean currents (gyres).
- Water, carbon, nitrogen and oxygen cycle through all spheres; human activity disturbs these cycles.
- Key terms: insolation, solar constant, albedo, urban heat island, valley/mountain breeze, planetary winds, ocean current, gyre, biogeochemical cycle, nitrogen fixation, nitrification, ammonification, denitrification, eutrophication.